Name: Cohort/Batch: Date: Score:

JUNIT PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Testing

TOTAL: 10 QUESTIONS

Q1 What is JUnit and what problem in Testing does it solve? Threat Model

Q2 What are the core components or concepts in JUnit? Architecture

Q3 How does JUnit compare to its main alternatives? Configuration

Q4 How do you get started with JUnit for a new project? Hardening

Q5 What configuration options most affect JUnit's behavior in production? SSO / IdP

Q6 How does JUnit handle reliability and high availability? Observability

Q7 What observability does JUnit provide out of the box? Lifecycle

Q8 What are common performance bottlenecks in JUnit? Compliance

Q9 What security best practices apply when running JUnit? Testing

Q10 What mistakes do teams commonly make when adopting JUnit? Tuning

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 **JUnit is a tool or platform that** Mitigation

JUnit is a tool or platform that automates or simplifies a specific concern in the Testing domain, reducing manual effort and operational complexity for engineering teams.

A2 **JUnit has key building blocks that work** Primitives

JUnit has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 **JUnit trades off simplicity against feature richness** Hardening

JUnit trades off simplicity against feature richness differently than its alternatives. Choose JUnit when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 **Install JUnit via its package manager or** Gotchas

Install JUnit via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 **Production-critical settings typically include concurrency** Integration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 **JUnit provides HA through leader election, replication,** Observability

JUnit provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 **JUnit exposes metrics (Prometheus endpoint or built-in** Automation

JUnit exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 **Performance bottlenecks typically stem from misconfigured** Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 **Run JUnit with the principle of least** Assessment

Run JUnit with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 **Common pitfalls include skipping the documentation and** Standardization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.